I.23 · Copy an angle by constructing a triangle

At a given straight line and a point on it, to construct a rectilinear angle equal to a given rectilinear angle.

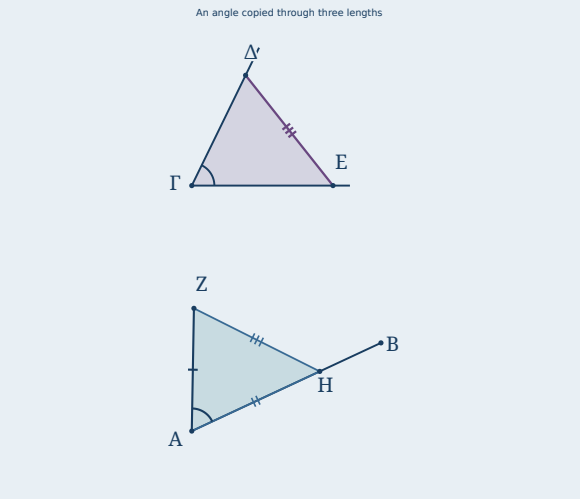


Plate at step 7: The requested angle has been made.

An angle copied through three lengths

Change the two point distances without changing the given angle. The auxiliary triangles change; the copied angle does not. Tilting AB changes the new ray's direction while retaining the relative opening. Both reflected copies have the same angular magnitude. Degrees and coordinates are modern illustrations; I.22 and I.8 supply the source's construction and proof.

Three side copies, then one angle equality

Given ΔΓE	Constructed ZAH	Correspondence
First side · ΓΔ / AZ	180.000	180.000
Second side · ΓE / AH	207.000	207.000
Closing side · ΔE / ZH	206.353	206.353
Included angle	64.000°	64.000°

`ΓΔ = AZ · ΓE = AH · ΔE = ZH · ∠ΔΓE = ∠ZAH (I.8)`

The proposition

Πρὸς τῇ δοθείσῃ εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῇ δοθείσῃ γωνίᾳ εὐθυγράμμῳ ἴσην γωνίαν εὐθύγραμμον συστήσασθαι.	Ad datam rectam et punctum in ea datum angulum rectilineum dato angulo rectilineo aequalem construere.	At a given straight line and a point on it, to construct a rectilinear angle equal to a given rectilinear angle.

1. A line, its point, and a separate angle Given

Ἔστω ἡ μὲν δοθεῖσα εὐθεῖα ἡ ΑΒ, τὸ δὲ πρὸς αὐτῇ σημεῖον τὸ Α, ἡ δὲ δοθεῖσα γωνία εὐθύγραμμος ἡ ὑπὸ ΔΓΕ· δεῖ δὴ πρὸς τῇ δοθείσῃ εὐθείᾳ τῇ ΑΒ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ Α τῇ δοθείσῃ γωνίᾳ εὐθυγράμμῳ τῇ ὑπὸ ΔΓΕ ἴσην γωνίαν εὐθύγραμμον συστήσασθαι.	Sit data recta AB et punctum in ea datum A et datus angulus rectilineus ΔΓΕ. oportet igitur ad datam rectam AB et punctum in ea datum A angulum rectilineum dato angulo rectilineo ΔΓΕ aequalem construere.	Let AB be the given straight line, A the point on it, and ΔΓΕ the given rectilinear angle. We must construct, at A on AB, a rectilinear angle equal to ΔΓΕ.

πρὸς with the dative locates the construction at a line and its point. δοθείσῃ εὐθείᾳ and δοθείσῃ γωνίᾳ have the same endings but different roles: one is the new support, the other the reference angle. Latin aequalem takes the dative dato angulo.

Reason: The task and its given objects A given object is part of the problem. It has not been constructed by this proof. A finite straight line here means a line segment.

Printed source: Greek pp. 56; Latin pp. 57.

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2. Choose any two points and join them Construct

Εἰλήφθω ἐφ’ ἑκατέρας τῶν ΓΔ, ΓΕ τυχόντα σημεῖα τὰ Δ, Ε, καὶ ἐπεζεύχθω ἡ ΔΕ·	sumantur in utraque ΓΔ, ΓΕ quaelibet puncta Δ, Ε et ducatur ΔΕ.	On each of ΓΔ and ΓΕ let arbitrary points Δ and Ε be taken, and let ΔΕ be joined.

τυχόντα / quaelibet means arbitrary: the two distances are not required to agree. Εἰλήφθω is a singular perfect passive imperative with neuter plural σημεῖα as subject, a familiar Greek agreement pattern; Latin sumantur is plural. ἐπεζεύχθω is another instruction, now for the joining line.

Reason: Postulate 1 · close the given angle into a triangle Choose a positive distance on each ray of the given angle, then join those points. Their distances need not be equal. This auxiliary triangle supplies three lengths; its angle at Γ remains the given angle.

Printed source: Greek pp. 56; Latin pp. 57.

3. Reuse the three-length construction Construct

καὶ ἐκ τριῶν εὐθειῶν, αἵ εἰσιν ἴσαι τρισὶ ταῖς ΓΔ, ΔΕ, ΓΕ, τρίγωνον συνεστάτω τὸ ΑΖΗ,	et ex tribus rectis, quae aequales sunt tribus rectis ΓΔ, ΔΕ, ΓΕ, triangulum construatur ΑΖΗ,	From three straight lines equal to ΓΔ, ΔΕ and ΓΕ, let triangle AZH be constructed...

ἐκ / ex introduces what the triangle is made from. τρισί and tribus are datives after the equality: equal to the three given lengths. συνεστάτω / construatur requests a completed triangle. The Latin's inline reference identifies I.22; the reader opens that earned result rather than treating it as a primitive operation.

Reason: I.22 · construct a triangle from those three lengths The three lengths already form triangle ΔΓΕ, so I.20 supplies all three strict triangle inequalities. I.22 can therefore construct AZH at the new point, with AH on the given ray. The construction places the second copied side, ΓΕ, along AB.

Printed source: Greek pp. 56; Latin pp. 57.

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4. Keep the side correspondence Construct

ὥστε ἴσην εἶναι τὴν μὲν ΓΔ τῇ ΑΖ, τὴν δὲ ΓΕ τῇ ΑΗ, καὶ ἔτι τὴν ΔΕ τῇ ΖΗ.	ita ut sit ΓΔ = ΑΖ, ΓΕ = ΑΗ, ΔΕ = ΖΗ.	...so that ΓΔ equals AZ, ΓΕ equals AH, and ΔΕ equals ZH.

ὥστε with εἶναι states the resulting configuration. μέν … δέ distinguishes the first two equalities, and καὶ ἔτι adds the third. The Latin writes the three equations explicitly. AH lies on the new line AB in the printed figure; our placement makes that required incidence explicit.

Reason: Keep the three side correspondences in order ΓΔ is copied as AZ, ΓΕ as AH, and ΔΕ as ZH. The order identifies the corresponding vertices: Γ ↔ A, Δ ↔ Z, E ↔ H. The equal-side marks show this relation without treating a degree reading as the construction.

Printed source: Greek pp. 56; Latin pp. 57.

5. Compare two sides and the closing base Prove

Ἐπεὶ οὖν δύο αἱ ΔΓ, ΓΕ δύο ταῖς ΖΑ, ΑΗ ἴσαι εἰσὶν ἑκατέρα ἑκατέρᾳ, καὶ βάσις ἡ ΔΕ βάσει τῇ ΖΗ ἴση,	iam quoniam duae rectae ΔΓ, ΓΕ duabus ΖΑ, ΑΗ aequales sunt altera alteri, et basis ΔΕ basi ΖΗ aequalis,	Since the two sides ΔΓ and ΓΕ equal ΖΑ and ΑΗ respectively, and base ΔΕ equals base ΖΗ…

ἑκατέρα ἑκατέρᾳ / altera alteri pairs each side with its counterpart. βάσις is nominative and βάσει dative: base ΔΕ is equal to base ΖΗ. The angle at Γ corresponds to the angle at Α; changing the names' order must not change the correspondence.

Reason: I.8 · three sides determine the included angle The two sides around Γ equal the two sides around Α, and their closing bases are also equal. I.8 proves the included angles equal. I.4 would require that angle equality as an input, so using it here would assume the desired conclusion.

Printed source: Greek pp. 56; Latin pp. 57.

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6. Infer equality of the included angles Prove

γωνία ἄρα ἡ ὑπὸ ΔΓΕ γωνίᾳ τῇ ὑπὸ ΖΑΗ ἐστιν ἴση.	erit ∠ ΔΓΕ = ΖΑΗ.	…the angle ΔΓΕ therefore equals the angle ΖΑΗ.

ἄρα marks an inference, not an additional construction. γωνία is the subject; γωνίᾳ is the dative comparison after ἴση. Heiberg's Latin cites I.8. It does not cite I.4, whose side–angle–side hypothesis would already require the angle equality we are proving.

Reason: I.8 · three sides determine the included angle The two sides around Γ equal the two sides around Α, and their closing bases are also equal. I.8 proves the included angles equal. I.4 would require that angle equality as an input, so using it here would assume the desired conclusion.

Printed source: Greek pp. 56; Latin pp. 57.

7. The requested angle has been made Conclude

Πρὸς ἄρα τῇ δοθείσῃ εὐθείᾳ τῇ ΑΒ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ Α τῇ δοθείσῃ γωνίᾳ εὐθυγράμμῳ τῇ ὑπὸ ΔΓΕ ἴση γωνία εὐθύγραμμος συνέσταται ἡ ὑπὸ ΖΑΗ· ὅπερ ἔδει ποιῆσαι.	Ergo ad datam rectam AB et punctum in ea datum A dato angulo rectilineo ΔΓΕ aequalis constructus est angulus rectilineus ZAH; quod oportebat fieri.	Thus angle ZAH has been constructed at A on AB, equal to the given rectilinear angle ΔΓΕ: which was required to be made.

συνέσταται announces the completed construction, answering the heading's συστήσασθαι. Latin constructus est and quod oportebat fieri likewise close a task of making. Equal angular magnitude does not require the two triangles to occupy the same position or orientation.

Reason: A new construction has been earned An angle equal to the given rectilinear angle has been made at the specified point on the given line. Earlier line-copy, triangle-construction and SSS results supplied the operation. A reflected triangle has the same angular magnitude; its orientation is a separate choice.

Printed source: Greek pp. 56; Latin pp. 57.

A construction becomes a more capable tool

The points Δ and E are arbitrary, so the three auxiliary lengths can change while the given angle stays fixed. Every successful side copy still gives the same opening. I.22 earns the triangle; I.8 earns the angle equality. Rotating the destination line changes its absolute direction, and reflecting the triangle changes its orientation, without changing that opening.

Definitions and starting permissions used in this proof

Postulate 1 · α′ / I · Join two points

Ἠιτήσθω ἀπὸ παντὸς σημείου ἐπὶ πᾶν σημεῖον εὐθεῖαν γραμμὴν ἀγαγεῖν.	Postuletur, ut a quouis puncto ad quoduis punctum recta linea ducatur.	Let it be postulated that a straight line can be drawn from any point to any point.

Ἠιτήσθω / Postuletur means 'let it be requested / postulated'. The Greek then uses an infinitive, ἀγαγεῖν ('to draw'); Latin uses ut … ducatur, 'that … be drawn'. The following postulates continue the opening request.

Printed source: Greek pp. 8; Latin pp. 9.

Greek and Latin: Euclidis opera omnia, vol. I, ed. and Latin translation J. L. Heiberg, Leipzig: Teubner, 1883, printed pp. 2–57. Public domain. University of Toronto scan. English, reading notes and redrawn coordinate diagrams: Stoicheia (2026), MIT. Greek ϑ is normalized to θ; line breaks and

inline references are omitted. Latin u/v spelling is retained. Heiberg’s square brackets in I.1, I.4, I.8, I.13 and I.15 are retained. Printed angle and triangle symbols are represented as ∠ and △. The English is a reading translation, including explicit explanations of implied intersections.

The diagram is numerical evidence, not a formal proof. Length units and orientation are modern teaching choices.

The printed evidence

Source pages from Heiberg’s 1883 edition, including the passages and foundations used in this saved reading. Every image and the bundled font were checked against this edition before saving. The apparatus is available for inspection; it is not fully transcribed. Where the original scan is cropped, the page caption identifies that limit.

οὔτε ὀρθογώνιον· τὰ δὲ παρὰ ταῦτα τετράπλευρα τραπέζια καλείσθω.

κγ΄. Παράλληλοί εἰσιν εὐθεῖαι, αἵτινες ἐν τῷ αὐτῷ ἐπιπέδῳ οὖσαι καὶ ἐκβαλλόμεναι εἰς ἄπειρον ἐφ' ἑκάτερα τὰ μέρη ἐπὶ μηδέτερα συμπίπτουσιν ἀλλήλαις.

Αἰτήματα.

α΄. Ἠιτήσθω ἀπὸ παντὸς σημείου ἐπὶ πᾶν σημεῖον εὐθεῖαν γραμμὴν ἀγαγεῖν.

β΄. Καὶ πεπερασμένην εὐθεῖαν κατὰ τὸ συνεχὲς ἐπ' εὐθείας ἐκβαλεῖν.

γ΄. Καὶ παντὶ κέντρῳ καὶ διαστήματι κύκλον γράφεσθαι.

δ΄. Καὶ πάσας τὰς ὀρθὰς γωνίας ἴσας ἀλλήλαις εἶναι.

ε΄. Καὶ ἐὰν εἰς δύο εὐθείας εὐθεῖα ἐμπίπτουσα τὰς ἐντὸς καὶ ἐπὶ τὰ αὐτὰ μέρη γωνίας δύο ὀρθῶν ἐλάσσονας ποιῇ, ἐκβαλλομένας τὰς δύο εὐθείας ἐπ' ἄπειρον συμπίπτειν, ἐφ' ἃ μέρη εἰσὶν αἱ τῶν δύο ὀρθῶν ἐλάσσονες.

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23. Hero def. 71. Philoponus in anal. II fol. 18^{v}. Psellus p. 35. Martianus Capella VI, 712. Boetius p. 376, 23. αἰτ. 1—5. Martianus Capella VI, 722. Boetius p. 377, 4. Aspasius apud Simplicium in Arist. de coelo fol. 149: τὰ πέντε αἰτήματα. 1. Philop. in anal. II fol. 9^{v}. 10. 29. 2. Simplicius in phys. fol. 119. 3. Philop. in anal. II fol. 10. 29. 4. Id. ibid. fol. 10. 5. Id. ib. fol. 10. 29. Proclus p. 364, 14.

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1. τετράγωνα B. 2. τραπέζεια b. Def. 21 uulgo in 3, def. 22 in 5 diuidunt. 3. παράλληλοι δέ B. εὐθεῖαί εἰσιν Proclus, Psellus. 4. ἐς V. 5. συμπίπτειν P. ἀλλήλαις om. F. 6. αἰτήματα πέντε V, αἰτ. ἐστι πέντε BF, b m. 2. Numeros om. F. 9. ἐπ' εὐθείας κατὰ τὸ συνεχές PBFbp$_1$

liqua autem praeter haec quadrilatera trapezia adpellentur.

XXIII. Parallelae sunt lineae, quae in eodem plano positae et in utramque partem productae in infinitum in neutra parte concurrunt.

Postulata.

I. Postuletur, ut a quouis puncto ad quoduis punctum recta linea ducatur.

II. Et ut recta linea terminata in directum educatur in continuum.

III. Et ut quouis centro radioque circulus describatur.

IV. Et omnes rectos angulos inter se aequales esse.

V. Et, si in duas lineas rectas recta incidens angulos interiores et ad eandem partem duobus rectis minores effecerit, rectas illas in infinitum productas concurrere ad eandem partem, in qua sint anguli duobus rectis minores.

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receptum ordinem tuentur V, Proclus, Simplicius, Capella, Boetius, Campanus. 10. *ἐκβάλλειν* V. 11. *γράφεσθαι*] codd. omnes et Philoponus; *γράψαι* ex Proclo recepit August. 13. *ἀλλήλαις*] om. V. 15. *εὐθειά τις* P. 17. *ἐλάττονας* Proclus p. 191, 18 (non p. 364). *τὰς δύο*] PBVbp, *δύο* om. F, Proclus bis, Martianus Capella, Boetius, fort. recte. 18. *συμπίπτειν τὰς εὐθείας ἐκβαλλομένας ἐφ'* Proclus p. 364. *συμπίπτειν ἀλλήλαις* PV (*ἀλλήλαις* corr. ex *ἀλλήλας* P). 19. *ἐλάσσονες*] Pp, Proclus p. 364; *ἐλάττονες* uulgo. Dein add. *γωνίαι* FBVb, Philoponus; om. Proclus bis et Pp. In ed. Basil. et apud Gregorium *αἴτ.* 4—5 inter communes notiones (10—11) leguntur (*πᾶσαι αἱ ὀρθαὶ γωνίαι ἴσαι .. εἰσί; ἐκβαλλόμεναι αἱ .. εὐθεῖαι .. συμπεσοῦνται*). Post *αἴτ.* 5 in PF et V m. 2 et apud Campanum sequitur: *καὶ δύο εὐθείας χωρίον μὴ περιέχειν.*

τὸ Δ ἄπειρος δὲ κατὰ τὸ Ε, καὶ κείσθω τῇ μὲν Α ἴση ἡ ΔΖ, τῇ δὲ Β ἴση ἡ ΖΗ, τῇ δὲ Γ ἴση ἡ ΗΘ· καὶ κέντρῳ μὲν τῷ Ζ, διαστήματι δὲ τῷ ΖΔ κύκλος γεγράφθω ὁ ΔΚΛ· πάλιν κέντρῳ μὲν τῷ Η, διαστήματι δὲ τῷ ΗΘ κύκλος γεγράφθω ὁ ΚΛΘ, καὶ ἐπεζεύχθωσαν αἱ ΚΖ, ΚΗ· λέγω, ὅτι ἐκ τριῶν εὐθειῶν τῶν ἴσων ταῖς Α, Β, Γ τρίγωνον συνέσταται τὸ ΚΖΗ.

Ἐπεὶ γὰρ τὸ Ζ σημεῖον κέντρον ἐστὶ τοῦ ΔΚΛ κύκλου, ἴση ἐστὶν ἡ ΖΔ τῇ ΖΚ· ἀλλὰ ἡ ΖΔ τῇ Α ἐστιν ἴση. καὶ ἡ ΚΖ ἄρα τῇ Α ἐστιν ἴση. πάλιν, ἐπεὶ τὸ Η σημεῖον κέντρον ἐστὶ τοῦ ΛΚΘ κύκλου, ἴση ἐστὶν ἡ ΗΘ τῇ ΗΚ· ἀλλὰ ἡ ΗΘ τῇ Γ ἐστιν ἴση· καὶ ἡ ΚΗ ἄρα τῇ Γ ἐστιν ἴση. ἐστὶ δὲ καὶ ἡ ΖΗ τῇ Β ἴση· αἱ τρεῖς ἄρα εὐθεῖαι αἱ ΚΖ, ΖΗ, ΗΚ τρισὶ ταῖς Α, Β, Γ ἴσαι εἰσίν.

Ἐκ τριῶν ἄρα εὐθειῶν τῶν ΚΖ, ΖΗ, ΗΚ, αἵ εἰσιν ἴσαι τρισὶ ταῖς δοθείσαις εὐθείαις ταῖς Α, Β, Γ, τρίγωνον συνέσταται τὸ ΚΖΗ· ὅπερ ἔδει ποιῆσαι.

κγ'.

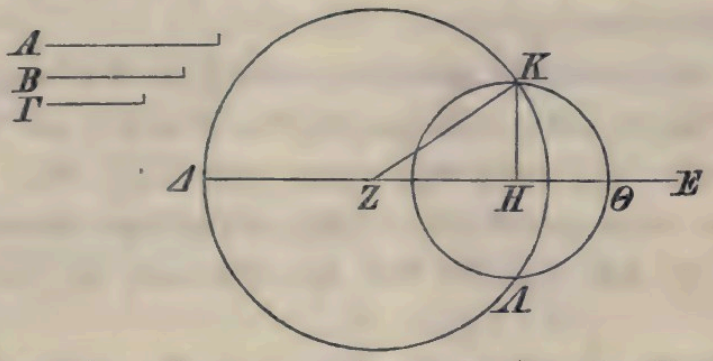
Πρὸς τῇ δοθείσῃ εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῇ δοθείσῃ γωνίᾳ εὐθυγράμμῳ ἴσην γωνίαν εὐθύγραμμον συστήσασθαι.

XXIII. Boetius p. 382, 5.

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1. τῇ] postea insertum m. 1 V. 2. ἡ] (tert.) m. rec. P. 3. μέν] om. b, Proclus. 4. καὶ πάλιν V, Proclus. μέν] om. V, Proclus. διαστήματι δέ] καὶ διαστήματι P. 7. συνέστηκε V; συνίσταται p. τό] corr. ex τῷ b. 8. γάρ] οὖν P. ἐστίν P. 9. ΖΔ] ΔΖ F. ἀλλ F. ΖΔ] ΔΖ V (ante Δ ras., Ζ mg. m. 2). 10. καὶ ἡ ΚΖ ἄρα τῇ Α ἐστιν ἴση] mg. m. 2 V. 11. ἐστίν Bb. ΔΚΘ] ΚΛΘ P, et in ras. V. 12. ἀλλ' F. 13. ΚΗ] corr. ex ΚΘ m. 2 P. 14. ΗΚ BF. ἐστιν ἴση] mg. m. 2 V. ἐστὶν δέ P. 16. τῶν]

tem infinita, et ponatur $\Delta Z = A$, $ZH = B$, $H\Theta = \Gamma$. et centro Z radio autem $Z\Delta$ circulus describatur $\Delta K\Lambda$. rursus centro H radio autem $H\Theta$ circulus describatur $K\Lambda\Theta$, et ducantur KZ, KH. dico, ex tribus rectis aequalibus rectis A, B, Γ triangulum constructum esse KZH.



nam quoniam Z punctum centrum est circuli $\Delta K\Lambda$, erit $Z\Delta = ZK$; uerum $Z\Delta = A$; quare etiam $KZ = A$ [κ. $\check{\epsilon}\nu\nu$. 1].[1]) rursus quoniam H punctum centrum est circuli $\Lambda K\Theta$, erit $H\Theta = HK$; uerum $H\Theta = \Gamma$; quare etiam $KH = \Gamma$. et praeterea $ZH = B$. itaque tres rectae KZ, ZH, HK tribus A, B, Γ aequales sunt.

Ergo ex tribus rectis KZ, ZH, HK, quae tribus datis rectis A, B, Γ aequales sunt, triangulus constructus est KZH; quod oportebat fieri.

XXIII.

Ad datam rectam et punctum in ea datum angulum rectilineum dato angulo rectilineo aequalem construere.

1) Cfr. Alexander Aphrod. in anal. I fol. 8. Studien p. 195.

τοῦ F. 17. τρισί] om. F. Γ] om. V. 18. συνίσταται p. 21. εὐθυγράμμῳ γωνίᾳ Proclus.

Ἔστω ἡ μὲν δοθεῖσα εὐθεῖα ἡ AB, τὸ δὲ πρὸς αὐτῇ σημεῖον τὸ A, ἡ δὲ δοθεῖσα γωνία εὐθύγραμμος ἡ ὑπὸ $\Delta\Gamma E$· δεῖ δὴ πρὸς τῇ δοθείσῃ εὐθείᾳ τῇ AB καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ A τῇ δοθείσῃ γωνίᾳ εὐθυγράμμῳ τῇ ὑπὸ $\Delta\Gamma E$ ἴσην γωνίαν εὐθύγραμμον συστήσασθαι.

Εἰλήφθω ἐφ' ἑκατέρας τῶν $\Gamma\Delta$, ΓE τυχόντα σημεῖα τὰ Δ, E, καὶ ἐπεζεύχθω ἡ ΔE· καὶ ἐκ τριῶν εὐθειῶν, αἵ εἰσιν ἴσαι τρισὶ ταῖς $\Gamma\Delta$, ΔE, ΓE, τρίγωνον συνεστάτω τὸ AZH, ὥστε ἴσην εἶναι τὴν μὲν $\Gamma\Delta$ τῇ AZ, τὴν δὲ ΓE τῇ AH, καὶ ἔτι τὴν ΔE τῇ ZH.

Ἐπεὶ οὖν δύο αἱ $\Delta\Gamma$, ΓE δύο ταῖς ZA, AH ἴσαι εἰσὶν ἑκατέρα ἑκατέρᾳ, καὶ βάσις ἡ ΔE βάσει τῇ ZH ἴση, γωνία ἄρα ἡ ὑπὸ $\Delta\Gamma E$ γωνίᾳ τῇ ὑπὸ ZAH ἐστιν ἴση.

Πρὸς ἄρα τῇ δοθείσῃ εὐθείᾳ τῇ AB καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ A τῇ δοθείσῃ γωνίᾳ εὐθυγράμμῳ τῇ ὑπὸ $\Delta\Gamma E$ ἴση γωνία εὐθύγραμμος συνέσταται ἡ ὑπὸ ZAH· ὅπερ ἔδει ποιῆσαι.

κδ'.

Ἐὰν δύο τρίγωνα τὰς δύο πλευρὰς [ταῖς] δύο πλευραῖς ἴσας ἔχῃ ἑκατέραν ἑκατέρᾳ, τὴν δὲ γωνίαν τῆς γωνίας μείζονα ἔχῃ τὴν ὑπὸ τῶν ἴσων εὐθειῶν περιεχομένην, καὶ τὴν βάσιν τῆς βάσεως μείζονα ἕξει.

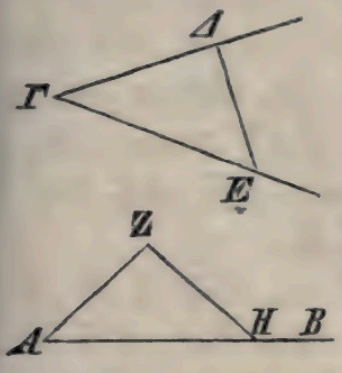
Ἔστω δύο τρίγωνα τὰ $AB\Gamma$, ΔEZ τὰς δύο πλευ-

XXIV. Boetius p. 382, 9.

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7. ἑκατέρα P. $\Delta\Gamma$ P. ΓE] eras. F. 9. Post ἴσαι

Sit data recta AB et punctum in ea datum A et datus angulus rectilineus $\Delta\Gamma E$. oportet igitur ad datam rectam AB et punctum in ea datum A angulum rectilineum dato angulo rectilineo $\Delta\Gamma E$ aequalem construere.



sumantur in utraque $\Gamma\Delta$, ΓE quaelibet puncta Δ, E et ducatur ΔE. et ex tribus rectis, quae aequales sunt tribus rectis $\Gamma\Delta$, ΔE, ΓE, triangulus construatur AZH, ita ut sit $\Gamma\Delta = AZ$, $\Gamma E = AH$ $\Delta E = ZH$ [prop. XXII].

iam quoniam duae rectae $\Delta\Gamma$, ΓE duabus ZA, AH aequales sunt altera alteri, et basis ΔE basi ZH aequalis, erit $\angle \Delta\Gamma E = ZAH$ [prop. VIII].

Ergo ad datam rectam AB et punctum in ea datum A dato angulo rectilineo $\Delta\Gamma E$ aequalis constructus est angulus rectilineus ZAH; quod oportebat fieri.

XXIV.

Si duo trianguli duo latera duobus lateribus aequalia habent alterum alteri et angulorum rectis aequalibus comprehensorum alterum altero maiorem habent, etiam basim basi maiorem habebunt.

Sint duo trianguli $AB\Gamma$, ΔEZ duo latera AB,

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add. V m. 2: *ταῖς δοθείσαις εὐθείαις.* *τρισίν* P. ΓE] mutat. in $E\Gamma$ V. 13. *δύο*] (alt.) *δυσί* FB. ZA] AZ F. 14. *ἑκατέρα*] supra m. 1 F. 15. *ἄρα*] m. 2 P. 19. *συνίσταται* p. 22. *τάς*] om. Proclus. *ταῖς*] om. Proclus. *δύο*] (alt.) P, Proclus; *δυσί* uulgo. 23. *ἔχῃ δὲ τὴν γωνίαν τῆς γωνίας μείζονα τήν* Proclus.